Attribution Functionalism:
A Strictly Functionalist Account of Phenomenal State Attribution

Mark Phelan
Associate Professor of Philosophy
Lawrence University

Abstract: A survey of the experimental philosophy of consciousness literature over the last decade makes one thing abundantly clear: people exploit functional information—information about the functions an entity can satisfy; i.e., the things it can do—in the process of attributing phenomenally conscious states to other entities. However, aspects of this literature also seem to suggest that people exploit certain pieces of non-functional information—information that does not contribute to functional information—in the process of attributing phenomenally conscious states to other entities. In this chapter, I argue, contrary to initial appearances, that compelling evidence for this second claim is not to be found in the experimental philosophy of mind. In fact, those studies that purport to supply such evidence suffer from a systematic failure to investigate and control for manipulations of functional information. Through a pair of experiments, I show how inferences to functional information are not controlled for in studies purporting to show a role for non-functional information in phenomenally conscious state attributions. This undercuts purported evidence against a strong functionalist account of phenomenal state attribution.

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We often think in functional terms about other objects, entities, and persons. According to prominent historical and ahistorical accounts of function,[1] when we do this we are thinking about the proper function that something (including some behavior) exists to produce (e.g., "its stripes function as a camouflage") or about the role something plays in a complex system (e.g., "it's using its paw like a cup, to bring water to its mouth"). Of course, we typically cannot directly perceive an object's function. So we must infer it, from other things we can perceive about the entity or the situation it finds itself in, or things we know about other entities of its kind, and so on. I know that many animals use stripes as camouflage, and so I infer that my tabby cat's tawny stripes exist because they provided camouflage to his ancestors, if not to him directly. I see an entity move out of the way of an object that is speeding towards it and I infer that the purpose of the entity's movement was avoidance. It is clear, then, that we infer functions from things we know about an entity and its situation, but it is unclear what specific aspects of knowledge constitute the base of inference to function or how such inferences proceed.

Not only do we often infer functional capacities for the entities we encounter, I take it that attributing functional capacities to an entity typically entails attributing certain causal relations involving psychological states to that entity. So, for example, inferring that some entity avoided an approaching object entails attributing certain causal relations involving the psychological states of that entity, such as "the object caused a perceptual experience in the entity," "the perceptual experience caused certain beliefs in the entity," and "the beliefs, together with pre-existing desires of the entity, caused it to behave in a way indicative of avoidance." If people readily infer functional capacities for the entities they encounter and if such inferences entail the attribution of causally implicated psychological states, then it seems plausible to suppose that

[1] See Millikan (1989) for a prominent example of a historical account of function. See Cummings (1975) for a prominent ahistorical account.

that this is a primary route by which people attribute psychological states to other entities. People attribute those psychological states they take to be entailed by the functional capacities they attribute to the relevant entity. Call this position on ordinary psychological state attribution *Attribution Functionalism*. The extreme version of this view would claim that functional information is the only information relevant to folk psychological state attribution. In this chapter, I will argue that the extreme version of Attribution Functionalism has not been ruled out, despite some seeming evidence to the contrary. I argue, on the basis of my own experimental results, that this purported evidence is plausibly explained by experimental participants' inferences to functional capacity, so that the results embodiment studies actually do nothing more than reiterate that people attribute those psychological states they take to be entailed by the functional capacities of a given entity.

I. Extreme Attribution Functionalism, the Embodiment Thesis, and Phenomenal States

It is often assumed within the experimental philosophy of mind literature, that something approaching the extreme version of attribution functionalism is correct for the class of psychological states that philosophers delineate as intentional states, which are characteristically regarded as non-phenomenal—e.g., willing, the having of beliefs, desires, and intentions, the having of concepts, and so on. However, some influential early articles in the experimental philosophy of mind (especially, Robbins and Jack, 2006, 2012, Gray et al. 2007, and Knobe and Prinz, 2008) suggest a different account for at least some of those psychological states that philosophers delineate as phenomenally conscious mental states—e.g., having the experience of seeing red, feeling pain, feeling happy, and so on. These articles argue that some information that does not contribute to functional information is relevant to folk phenomenal state attributions, even if it is not relevant to folk non-phenomenal, intentional state attributions. In this section, I'll review some of this literature.

It is clear from the last ten-plus years of the experimental philosophy of mind—as well as from the previous decades of experimental psychology—that people exploit functional information in deciding what phenomenally conscious states to attribute to other entities. A partial survey of results from experimental philosophy is sufficient to establish this conclusion:

- Sytsma (2010) found that a pair of conjoined twins who "grimaced and shouted out 'Ouch!' after they "forcefully kicked a large rock" with their shared leg, were assessed by experimental participants as clearly sharing the same pain.
- Sytsma and Machery (2010) found that a simple robot, which lacks typical sense organs and a brain, is thought by participants to be capable of seeing red and smelling a chemical when it discriminates a red box or a chemical-scented box from other boxes.
- Buckwalter and Phelan (2013) found that a simple robot is assessed as more capable of occupying a range of experiential states (e.g., smelling bananas, feeling guilt) when it is described as having been designed for a certain, functional purpose (e.g., to make smoothies, to be an emotional support to the elderly) and as behaving in ways indicative of a certain functional capacity (e.g., as discriminating bananas when exposed only to their smell, as avoiding the target of guilt). A simple robot who lacks the relevant functional design or behaves in contra-functionally appropriate ways was assessed as less-capable of these experiential states.
- Buckwalter and Phelan (2014) found that embodied humans and disembodied spirits were attributed emotional states (e.g., feeling sadness, feeling happiness) at the same high rates when described as behaving in ways indicative of certain functional capacities (e.g., behaving in a vindictive way towards an ex, protecting a home from encroachers), and not when they behaved in contra-functionally appropriate ways.
- Salomons et al. (2021) found that participants agreed with an attribution of pain to a hospital patient who "moans, grimaces, cries out, clutches his stomach and, if asked, says that he is in severe pain" even in the absence of identifiable tissue damage. Furthermore, they found that people disagreed with an attribution of pain to a figure skater who damaged her wrist but didn't behave in ways associated with pain.

Let's refer to these studies and studies like them collectively as *function studies*. In a function study, some phenomenal state (e.g., experiencing pain, feeling an emotion) is attributed (or attributed at a higher rate) when participants are provided with information that the entity behaves (or can behave) in ways that are taken to be causally related to the relevant phenomenal state (e.g., favoring an injured body part, behaving vindictively towards an ex). And, in the last three function experiments I mentioned, phenomenal states are shown not to be attributed (or to be attributed at a much lower rate) when the sorts of things the entity is portrayed as doing or being capable of doing are not associated with the relevant phenomenal state. Clearly, then, people exploit functional information in deciding what phenomenally conscious states to attribute to other entities. Of course, this is unsurprising, since we often seem to attribute phenomenal states to other people and animals based on our observations of their behavior (e.g., she is shouting, he is crying) and our assumptions about their functional design (e.g., she is normally sighted, dogs get unhappy when they're hungry).

While it is clear, then, that functional information plays a role in the phenomenal state attribution process, some experimental philosophers suggest—contrary to an extreme attribution functionalist perspective—that exhibiting the appropriate functional profile is not, by itself, a decisive condition for lay-attributions of phenomenal states. These theorists maintain that at least some psychological states that would be characterized as phenomenally conscious mental states by philosophers of mind are attributed by (at least a portion of) lay people only to—or, at least, more enthusiastically to—entities that are embodied in a certain way. Specifically, these theorists maintain that lay people attribute phenomenal states (or some subset thereof) more readily to entities that possess a unified, biological body, than to entities that lack such a body, even if the two entities are assessed as functionally equivalent. It is a corollary of this position that (all else being equal) even if an entity is assessed as capable of the associated functional profile, attributions of phenomenal states will be withheld or less enthusiastically endorsed if the entity lacks a unified, biological body. Entities that might instantiate some relevant functional profile or other, but lack a unified, biological body include disembodied spirits (which lack a body), robots (which lack a biological body), and groups of people or swarms of insects (which lack a unified body). In fact, several influential studies have shown that endorsement rates for phenomenal state attributions to such entities were significantly lower than endorsement rates for non-phenomenal, intentional state attributions to those entities, or than attributions of phenomenal states to biologically embodied, functionally equivalent entities. These studies include:

- Gray, Gray, and Wegner (2007) found that a robot and God were assessed as less capable of experiential states (such as feeling pain or experiencing pride) than a host of other, biologically embodied entities (such as a frog, an infant, or an adult woman).
- Knobe and Prinz (2008) showed that attributions of phenomenal states (such as experiencing joy or feeling pain) to a fictional Acme Corporation are judged much "weirder" than attributions of non-phenomenal, intentional states (such as believing or intending) to that corporation.
- Huebner (2010) found that participants agreed strongly with attributions of non-phenomenal, belief states to a human, a robot, a robot with a human brain, or a human with a cpu instead of a brain. However, these participants were significantly more likely to agree with attributions of pain and happiness to the human than to the robot or either cyborg. Even though all four agents were said in Huebner's vignettes "to behave in every respect like a person on all psychological tests" (138).
- Sytsma (2012) found that while participants agreed with attributions of pain to an agent who burns her bioengineered, fleshy prosthetic hand, they somewhat disagreed with attributions of pain to an agent who burns her robotic, synthetic prosthetic hand, despite the fact that, in either case, the agent is said to be "able to do all of the things with her…[prosthetic]…hand that she did with her old hand" (190).
- McLaughlin and Rose (2018) showed that participants rejected attributions of feelings of pain, itches, warmth, and pressure to a lifelike robot neonate. And they showed this for a sample in which they removed from analysis any participant who rejected one or more of three comprehension statements: 1) Jane is a normal human neonate; 2) Jane-R is a robot neonate created by Kymoto; 3) Jane-R and Jane are indistinguishable in outward appearance and outward behavior.

Let's refer to these studies and studies like them collectively as *embodiment studies*. In an embodiment study, some phenomenal state (e.g., experiencing pain, feeling an emotion) is attributed (or attributed at a higher rate) to an entity that is described as being biologically embodied as compared with an entity that is not biologically embodied, even when participants are provided with information that the two entities do (or can) behave in the same ways. Not all of these studies are taken by their authors to support the sort of embodiment thesis discussed above. However, each has been taken to support embodiment by proponents of that thesis, as will become clearer in the discussion to follow. What's more, the specific form of embodiment to which a theorist is committed can vary. One embodiment theorist might maintain that having the appropriate sort of body is a necessary condition for being assessed as occupying some phenomenal state, whereas another might maintain that having the appropriate sort of body only increases the strength or frequency of such assessments. But all embodiment theorists are united in claiming that information about the body of an entity factors into (at least some lay people's) assessments of that entity's capacity for phenomenal states, and that it does so independently of functional information. This is the *Embodiment Thesis* that is at issue in this chapter. It is clearly incompatible with an extreme attribution functionalist view.

As I'll explain in section II, the embodiment studies have been taken to support the view that information about embodiment plays a role in phenomenal state attribution that is independent of the role played by functional information. However, as I'll argue in this paper, inferences to functional information are not controlled for in the embodiment studies. This undercuts purported evidence in favor of a role for non-functional information in the process of phenomenally conscious state attribution.

II. Embodiment Studies and the Embodiment Thesis

The embodiment studies have been used in support of various embodiment theses. Indeed, Gray et al. (2007) use their original comparison study to argue that we have "not one dimension of mind perception, but two," (619). In this early study, a robot and God were assessed as less capable of 'experiential' states than were entities such as a little girl and the family dog. God was judged as highly capable of 'agentic' states, however, so the seeds of a biological embodiment view for phenomenal state attribution were planted. Knobe and Prinz (2008) are more explicit in concluding that their experiments support a two-part hypothesis, according to which, if people classify a mental state ascription as one that requires phenomenal consciousness, they "will regard it as acceptable only if they feel that the mental state in question has the right sorts of physical constitution" (75). In other instances, the authors of the studies don't, themselves, explicitly argue for a bifurcated view of mental state ascriptions, according to which considerations of body factor into phenomenal, but not non-phenomenal state ascriptions. Huebner (2010), for example, only goes so far as to claim that, "while behavioral and functional considerations may be enough to warrant the ascription of belief…some entities also tend to be seen as possessing capacities for subjective experience that are not captured by the behavioral and functional properties of an entity" (144). But he stops short of endorsing a full-blown version of embodiment, reasoning that "the ambivalence about cyborg pain [revealed in his studies] may be driven by a corresponding ambivalence about the possibility of affective sensations in cyborgs" (141) and that "commonsense understanding of the mind is non-committal for many of the most philosophically interesting cases of mental state ascription" (143). Despite Huebner's caution on this front, his results have been widely taken to support embodiment. Knobe et al. (2012), for example, count them among a "diverse range of data [that] suggests that the attribution of phenomenal states is driven by different cues from the attribution of nonphenomenal states" (95).[2]

The embodiment studies, then, and studies like them are used in support of various embodiment theses. However, should they be used in this way? Do these sorts of studies actually support embodiment? I will argue that they do not because they suffer from a common trait that

[2] Elsewhere, Knobe et al. (2012) suggest that "complex behavior is the main cue for attributions of nonphenomenal states whereas embodiment is the main cue for attributions of phenomenal consciousness" (94).

undermines their ability to support an embodiment theory. Embodiment theories of phenomenal state attribution all essentially claim that information about an entity's body factors in ordinary phenomenal state attributions in a way that is independent of functional information. However, none of the relevant studies establish that their results arise independently of manipulations of functional information. We lack a systematic understanding of what sorts of manipulations lead lay people to infer information about the things an entity can do. In the absence of such information, those who would establish that considerations of body factor in phenomenal state attributions independently of functional information must control for participants' inferences about functional information, or else they must demonstrate that effects related to manipulations of body are not mediated by participants' inferences about function. Many of the embodiment studies attempt to control for inferences to functional information. However, as I will demonstrate through a series of experiments, these attempts are unsuccessful. Therefore, proponents of embodiment haven't ruled out an alternate explanation for the Embodiment Studies, that their results are due to subtle manipulations of participants' inferences to function.

III. Inferring Functional Information

What sorts of cues lead people to infer functional information about an entity? The function studies suggest some such cues. By depicting an entity as behaving in ways stereotypically associated with discriminating a particular color or avoiding a particular target, experimental philosophers seem to have been able to get participants to infer that the entity functions as a color-discriminator or a target-avoider and to ascent to phenomenal state attributions related to those functions more wholeheartedly. By describing an entity as being designed for a certain phenomenal-state-involving purpose, experimental philosophers were able to get participants to ascent to phenomenal state attributions related to that function more wholeheartedly. So, it seems that people infer functional information from information about the complex behaviors an entity performs and information about its design history. But, in addition to complex behavior and design history, what other sorts of cues lead people to infer function?

The psychological literature on how people represent function is instructive. This literature has developed with the primary aim of understanding object categorization. It has tended, therefore, to emphasize a unique, proper function for an object and to investigate the role this function plays in categorizations of the object. Several types of approaches to these topics are discernible. Following Gibson (1950, 1979) in understanding an affordance as a way an object can be used given its physical structure and the physical structure of the perceiving agent, some researchers have shown that young children prioritize functional affordances in object categorization (e.g., Kemler-Nelson et al., 2000; Oakes and Madole, 2008; Baumgartner and Oakes, 2011). Other researchers have shown that children emphasize a creator's original design intentions in determining an object's categorization-relevant function (e.g., Bloom, 1996, 1998; Gelman and Bloom, 2000). Other researchers, proponents of the 'HIPE' Theory, have promoted a maximalist view, according to which people's representations of an object's function are representations of a complex, relational structure comprising (potentially) facts about the object's history, its physical environment, and various of its behaviors and their outcomes (e.g., Chaigneau et al., 2004; Barsalou et al., 2005; Chaigneau et al., 2008). Importantly, this last view construes function as perspective relative, so it admits of distinct functions for a single object, depending on the observer's point of view and meta-cognitive needs at the time. Additionally, Chaigneau et al., 2004, demonstrates how distant features, such as an object's design history and an agent's goal in using it, can lead to inferences about an object's function in the absence of more immediate features, such as details about its physical structure and an agent's behaviors in using it. If the HIPE theory correctly describes ordinary representations of function, then inferences to an objects function may arise from knowledge of features as diverse as the object's creator's intentions, the object's particular life history, the setting in which the object appears, and the outcomes the object produces during use. Taken together, the psychological literature gives us many leads on the sorts of cues that may inform inferences to function. In the experiments that follow, I will primarily be investigating how an entity's personal history and the setting in which it appears play a role in inference to function.

IV. Study #1: Functional differences in Huebner's Cyborg Vignettes

Among the embodiment studies, most critical attention has been paid to Gray, Gray, and Wegner (2007) and Knobe and Prinz (2008).[3] But Huebner (2010) presents an interesting pair of studies that are often referenced in defense of embodiment theses, but which have not faced the critical scrutiny of the earlier studies. As we will see, these studies present an excellent illustration of how attempts to control for functional information can fall short. Once we control for inferences to function, the purported embodiment effects disappear.

In his studies, Huebner asked people about the psychological states of four types of entities: a normal human, a human whose brain has been replaced with a CPU, a normal robot with a CPU, and a robot whose CPU has been replaced by a brain. Here is Huebner's vignette for the human whose brain has been replaced by a CPU:

> **CPU Replacement:** This is a picture of David. David looks like a human. However, he has taken part in an experiment over the past year in which his brain has been replaced, neuron for neuron, with microchips that behave exactly like neurons. He now has a CPU instead of a brain. He has continued to behave in every respect like a person on all psychological tests throughout this change.

[3] See Phelan (in press) for an overview of these two works and responses to them.

The vignettes were paired with a picture of either a human or a humanoid robot, depending on the condition. After they had read the vignette, participants reported their level of agreement with the following mental state attribution sentences, to whichever agent they read about:

Study #1: He believes 2+2=4.
He feels pain if he is injured or damaged in some way.

Study #2: He believes that triangles have three sides.
He feels happy when he gets what he wants.

Participants recorded their responses on 5-point scales anchored at 1-strongly disagree and 5-strongly agree. Huebner performed analyses of variance (ANOVAs) on these results. However, citing concerns about treating Likert scale data as continuous data, he also converted participant responses into three-categories: disagree (comprising 'strongly disagree' and 'disagree' responses), agree (comprising 'strongly agree' and 'agree' responses), and ambivalent (comprising 'neither agree nor disagree' responses). Huebner performed Chi-square tests on this nominal data. Huebner found significant results for each method of analysis and across both studies. For example, univariate ANOVAs revealed significant differences between assessments of pain attributions ($p<0.001$) for Study 1 and happiness attributions ($p=0.005$) for Study 2 to Huebner's four entities. Bonferroni post-hoc tests revealed significant differences between the normal human and every other entity in Study 1, such that participants agreed with attributions of pain upon injury to the normal human more than to any other entity. Bonferroni post-hoc tests also revealed significant differences between the normal human, on the one hand, and both the robot with a CPU and the human with a CPU in Study 2, such that participants agreed with attributions of happiness upon satisfaction of desires to the normal human more than to either other entity. However, no significant differences emerged for attributions of belief in either study. Instead, participants registered high levels of agreement with belief attributions to each entity in both studies.

Huebner's studies were explicitly designed to "test the extent to which commonsense psychology relied on structural cues in evaluating the ascription of mental states to entities that were functionally equivalent" (137). And he goes on to infer from his results that:

> Commonsense ascriptions of happiness and pain introduce complications for anyone who defends the intuitive plausibility of functionalism. While behavioral and functional considerations may be enough to warrant the ascription of belief…some entities also tend to be seen as possessing capacities for subjective experience that are not captured by the behavioral and functional properties (144).

As mentioned previously, Huebner stops short of endorsing embodiment, merely arguing against functionalism for phenomenal state attribution; but his studies have been incorporated into arguments for embodiment over extreme attribution functionalism.]However, these studies constitute evidence against attribution functionalism only if people really see different of Huebner's entities as functionally equivalent. In his studies, Huebner attempts to ensure that this is the case by concluding each of his vignettes by specifying that the entity behaves "in every respect like a person on all psychological tests." However, as a standardization of function, this leaves much to be desired. For one thing, it presupposes on behalf of lay participants a uniform and quiet robust understanding of psychological testing. Contra this presupposition, it seems plausible that some people might suppose that people possess some functional capacities not captured by psychological testing. Indeed, as my first study shows, this is not an adequate control for inferences to function.

Method

Participants: A power analysis using G*Power revealed that at least 273 participants were needed to detect an effect of size f = 0.25 (d = 0.5), which is similar to some of the effects reported in Huebner (2010). Because Huebner's original sample sizes had been quite small, at around twenty participants per cell, and because my total university funding for my planned studies allowed it, I aimed to recruit at least 400 participants. In total, 422 participants (59% men, age M = 36.69, SD = 10.27) from Amazon's Mechanical Turk completed the study before it stopped receiving new HITs. Recruitment was not limited to Master MTurkers for this study, and participants were paid \$0.40 for participating. I preregistered to exclude participants who did not complete the study questions (though not the demographic probes), as well as the later participating of any duplicate IP addresses. In total, 15 participants were excluded for these reasons, leaving a sample size of 407.

Procedure: In previous research, Buckwalter and Phelan (2013) showed that participants were willing to attribute various olfactory experiences and emotional states to a simple robot, so long as it was described as performing certain function-related behaviors and being designed for certain phenomenal state involving purposes. Therefore, in this study, I focused on the differences Huebner uncovered between the normal human and the cyborg human who had his brain replaced with a CPU. Importantly, Huebner uncovered significant differences between participants' assessments of pain and happiness attributions for these two entities, such that the cyborg was assessed as less capable of both pain and happiness, perhaps due to its not wholly biological constitution. I described these two vignettes in the same way as Huebner and, like Huebner, I asked participants to assess their agreement on a five-point scale with the following three claims about David: 1) He believes that 2+2=4; 2) He feels pain if he is injured or damaged in some way; and 3) He feels happy when he gets what he wants.

The primary aim of my first study is to examine whether people who are told that an entity "behaves like a person on all psychological tests" actually regard that entity as functionally equivalent to a normal person, as Huebner (2010) presupposes. To do this, I added a probe to my vignettes, which asked participants' level of agreement with the statement, 4) He behaves like a normal person in every way. I am presupposing that lay participants have a concept of a 'normal person' and I take it that if a person is functionally equivalent to a normal person, then they will be regarded as behaving like a normal person in every way.[4] Further, this probe seems to capture what is at issue in the debate over embodiment, since the embodiment theorist wants to find evidence that information about bodies plays some role in phenomenal state attribution that is independent of the role played by functional information. Additionally, this method allows us to select those participants who *actually do* see the person whose brain has been replaced by a CPU as behaviorally equivalent to a normal person (so, as at least somewhat functionally equivalent), and to assess the degree to which biological embodiment plays a role in their phenomenal state attributions.

In addition to this primary modification of Huebner's method, I made two other modifications. First, I have long been struck by some, I think unintentional, differences between Huebner's belief probes and his phenomenal state probes. On the one hand, believing that 2+2=4 and that triangles have three sides are quite basic intentional states. It would be trivial to determine if someone held these beliefs through simple questioning. And it is possible to hold these beliefs non-occurrently—that is, someone can be said to believe something even if they're not thinking that thought in the moment. On the other hand, feeling pain if one is injured or damaged in some way and feeling happy when one gets what one wants are more complicated states. It would require repeated observation in different sorts of situations to determine if

[4] Note that the converse is not true; even if an entity behaves like a normal person in every way, it is not necessarily functionally equivalent to a normal person, since an entity might instantiate all relationships between sensory inputs and behavioral outputs that a normal person does, but via radically different interrelations between its mental states.

someone was subject to these sorts of dispositions. And feeling pain and happiness are occurrent states—that is, someone is said to feel pain or happiness only if they're feeling those states in the moment. Perhaps the effects Huebner uncovered are due to these differences—the triviality of possessing an intentional state such as believing that 2+2=4, for example—and not the difference that is at issue, between phenomenal and non-phenomenal states. To investigate this, I added a non-phenomenal, intentional state probe that more closely mirrored the phenomenal probes. I asked participants to assess their agreement with 5) He forms beliefs about items in his immediate vicinity when he looks around a room. This, like the states of pain and happiness queried in Huebner's probes, is a disposition to the occurrent state of forming beliefs, yet, unlike those other states, it is non-phenomenal.

Finally, I added a third condition to this first study, involving a new vignette, not originally investigated by Huebner. Phelan and Buckwalter (2012) discussed how there is a large difference in the personal history of normal human David and the David whose brain has been replaced, neuron by neuron, with microchips constituting a CPU.[5] Cyborg David is and normal David is not described as having gone through a time intensive and invasive experimental procedure. Is it possible that, as the HIPE model would suggest, participants are inferring that cyborg David's functional capacities have been shaped—perhaps stunted—by this procedure, and that this aspect of his personal history, not his partial non-biological embodiment, is why they judge him less capable of happiness and pain? To investigate this question, I asked participants about a third vignette, which described David as having undergone the same basic procedure, but having had his brain replaced, neuron for neuron, with other neurons:

[5] For reference, the normal human vignette, in Huebner (2010) and the present study, read as follows:

> **Normal Human:** This is a picture of David. David looks like a human and he has a normal human brain. He behaves in every respect like a person on all psychological tests.

> **Brain Replacement:** This is a picture of David. David looks like a human. However, he has taken part in an experiment over the past year in which his brain has been replaced, neuron for neuron, with biological, lab-grown neurons that behave exactly like his original neurons. He now has a fully regenerated brain. He has continued to behave in every respect like a person on all psychological tests throughout this change.

By including this third condition, we are able to isolate the distinction between biological embodiment and personal history. If the effects Huebner uncovered are due to biological embodiment as opposed to accidental features of David's personal history, then responses to this new condition should pattern more closely with Huebner's normal human probe than with his CPU replacement probe. If they are due, instead, to David's personal history, then responses to this new condition should pattern more closely with Huebner's CPU replacement probe than his normal human probe.

To summarize, in this mixed model 3x5 (entity, state) study, participants were randomly presented with one of three conditions involving three sorts of entities: a normal human, a human whose brain had been replaced with a CPU, and a human whose brain had been replaced with a new brain. Participants assessed their agreement with the five probes described above on 5-point scales anchored at 1-strongly disagree and 5-strongly agree. The order of the five probes was randomized. The presentation order did not meaningfully change the results, so I report the overall results below.

Results and Discussion

Analyses of all participants: To begin with, I analyzed results for all 407 participants who completed all five probes, regardless of responses to the *behaves* probe. One-way ANOVAs revealed no effect for entity on the *believes*, *forms beliefs*, or *feels happy* probes. However, univariate ANOVAs revealed small significant effects for entity on the *pain* and *behaves* probes. See Table 1 for details.

Probe	Normal Human *M*	Human Brain Replace *M*	Human CPU Replace *M*	F-score, η2
Believes: He believes that 2+2=4.	4.82	4.77	4.79	F (2, 404)=0.38, p=0.684, ηp2=0.002
Forms beliefs: He forms beliefs about items in his immediate vicinity when he looks around a room.	4.1	4.05	4.22	F (2, 404)=1.6, p=0.204, ηp2=0.008
Feels happy: He feels happy when he gets what he wants.	4.35	4.23	4.16	F (2, 404)=2.24, p=0.108, ηp2=0.011
Feels pain: He feels pain if he is injured or damaged in some way.	4.24	* 4.1	3.96	F (2, 404)=3.11, p=0.046, ηp2=0.015
Behaves: He behaves like a normal person in every way.	4.35	* 4.05	4.08	F (2, 404)=4.22, p=0.015, ηp2=0.020

Table 1: Results of One-way ANOVAs including all participants for five State probes and three Entities, Study 1.

Following Huebner, I conducted post-hoc Bonferroni tests, which revealed significant differences for normal human vs. human whose brain had been replaced by a CPU for *feels pain* (p = 0.039) and *behaves* (p = 0.017). No other significant results were found for Bonferroni tests and only two results had a p-value less than 0.2. Attributions of *feels happy* to a human whose brain had been replace by a CPU trended lower than attributions of *feels happy* for a normal homan (p = 0.112). Attributions of *behaves* to a person whose brain had been replace by other neurons trended lower than attributions of *behaves* for a normal person (p = 0.117).

Like Huebner, I converted participant responses into the three-categories of disagree (no), agree (yes), and ambivalent (not sure) and performed Chi-square tests on this nominal data. Specifically, Chi-square tests were performed for each of the states I probed to compare normal human with each transplant patient and the transplant patients with each other. Participants were significantly more likely to agree with attributions of *feels happy*, $X^2(2, N=275) = 13.318$, p = 0.001, and *behaves*, $X^2(2, N=275) = 9.062$, $p = 0.011$, to the normal person compared with the person whose brain had been replaced by a CPU.[6] They were also significantly more likely to agree with attributions of *feels happy*, $X^2(2, N=266) = 6.542$, $p = 0.038$, to the person whose brain had been replaced with other neurons than to the person whose brain had been replaced by a CPU.[7] Marginal differences were found between the normal person and the person whose brain had been replaced with other neurons, such that participants agreed more with attributions of *feels happy*, $X^2(2, N=273) = 5.75$, $p = 0.056$, *feels pain,* $X^2(2, N=273) = 4.936$, $p = 0.085$, and *behaves,* $X^2(2, N=273) = 4.709$, $p = 0.095$, to the normal person than to the person whose brain had been replaced by other neurons.[8] No other significant or marginally significant differences were found (see figure 1 for patterns of attribution agreement).

[6] Though, the chi-square test's assumption that the expected frequency for each cell should be greater than five was violated by one cell (4.87) for *feels happy*, threatening the accuracy of the statistic.
[7] Though, the chi-square test's assumption that the expected frequency for each cell should be greater than five was violated by one cell (4.96) for this result.
[8] Though, the chi-square test's assumption that the expected frequency for each cell should be greater than five was violated by two cells (minimum count, 0.97) for the *feels happy* result and by two cells (minimum count, 2.42) for the *behaves* result.

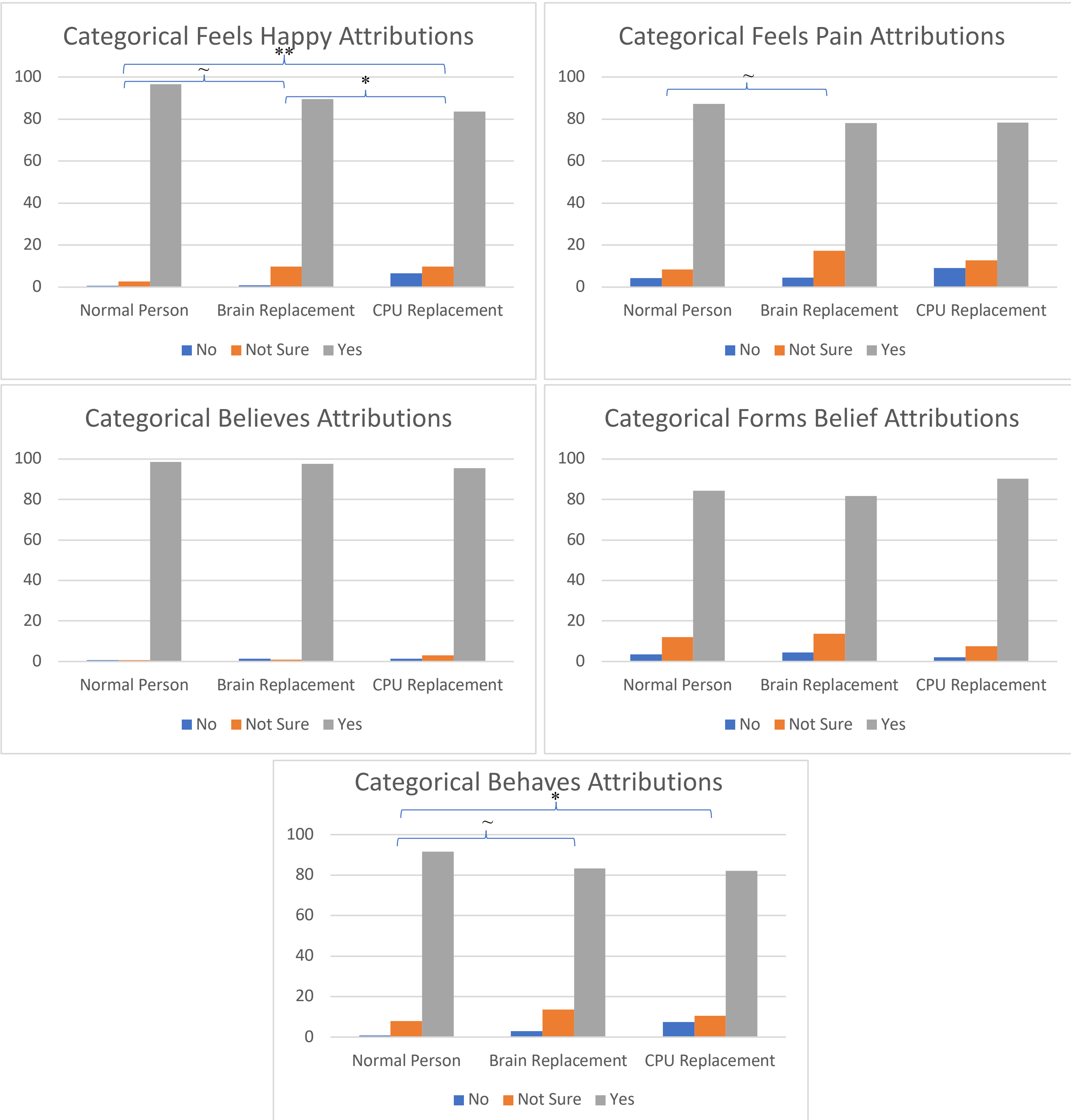

Figure 1: Categorical assessments of appropriateness for different state attributions, expressed as a percentile of 100% of answers for entity (analysis including all participants).

These results reflect Huebner's (2010) original findings in certain important respects. As with Huebner, an ANOVA failed to reveal any significant difference between ascriptions of belief to the various entities. Indeed, this was also the case with the *forms beliefs* probe; though, interestingly, there was a non-significant trend such that people agreed with attributions of belief formation *more* for the person whose brain had been replaced by a CPU than to any other entity. However, and again replicating Huebner (2010), there was a significant difference for attributions of pain to the various entities and post-hoc comparisons showed that participants were significantly less likely to attribute pain to the person with a CPU than to the normal person. Chi-square tests on the categorical version of the data also showed that people were significantly less likely to agree with attributions of happiness to the human with a CPU when compared with either the normal human or the human whose brain had been replaced with other neurons, providing some solace to the proponent of embodiment.

However, the current results also depart from Huebner in ways that should give the embodiment theorist pause. For one thing, marginal results for the categorical data suggest that people think of the person whose brain has been replaced by different neurons as quite psychologically distinct from a normal person—as less capable of feeling happiness and pain and as behaving in different ways than a normal person. If Huebner's original results for happiness and pain for the person whose brain had been replaced by a CPU were due to that entity's unique embodiment, then we would expect the person whose brain has been rebuilt with different neurons to pattern more closely with the normal person. Instead, patterns of agreement with attributions of phenomenal states to the two surgical replacement patients trended in the same direction, suggesting these results may have more to do with the surgical intervention these entities endured than their physical composition. What's more, participants rated attributions of the capacity to form beliefs much lower for all three entities ($M = 4.12$, $SD = 0.772$) than they did attributions of maintaining simple beliefs, such as 2+2=4 ($M = 4.79$, $SD = 0.545$). A paired samples T-test revealed this difference to be significant, $t(406) = 15.905$, $p < 0.001$. This result makes sense when we suppose that people are more likely to attribute the intentional state of believing 2+2=4 because of its utter triviality. But this explanation undercuts the embodiment theorist's argument from Huebner's original results, since feeling pain and happiness are non-trivial in the same ways as is forming beliefs about one's surroundings.

Finally, it's clear that both the person whose brain has been replaced by a CPU and the person whose brain has been rebuilt with other neurons are regarded by many as behaviorally—and, therefore, functionally—distinct from a normal person. But the embodiment theorists claimed that considerations of an entities body factored independently in people's attributions of phenomenal states. Is it, instead, just inferences to functional differences from non-functional factors that explain Huebner's results? To answer this question, I repeated the above analyses on just those participants who agreed that the entity they assessed behaved like a normal person in every way.

Analyses of participants who regard their target entity as behaving like a normal person: Eliminating participants who were unsure or unconvinced that the entity they rated behaved like a normal person—that is, eliminating participants who chose a response lower than four on the five-point Likert scale for the *Behaves* probe—left 349 participants. As mentioned previously, a proportionally greater number of participants agreed that the normal human behaved like a normal person than agreed with that assessment for either the human whose brain had been replaced by a CPU or the person whose brain had been rebuilt with other neurons. The numbers of participants in each condition of this reduced sample are included in Table 2:

Condition	*N* 2nd analyses (1st)
Normal Person	129 (141)
Neurons Transplant	110 (132)
CPU Transplant	110 (134)

Table 2: Numbers per cell, 2nd and 1st analyses.

One-way ANOVAs conducted on this population revealed no effect for entity on any of the remaining four probes *believes*, *forms beliefs*, *feels happy*, or *feels pain* (responses to *behaves* were used as a selection criterion for this sample, so analyses were not conducted for this probe). See Table 3 for details:

Probe	Normal Human *M*	Human Brain Replace *M*	Human CPU Replace *M*	F-score, η2
Believes: He believes that 2+2=4.	4.87	4.82	4.81	F (2, 346)=0.735, p=0.480, ηp2=0.004
Forms beliefs: He forms beliefs about items in his immediate vicinity when he looks around a room.	4.14	4.18	4.29	F (2, 346)=1.373, p=0.255, ηp2=0.008
Feels happy: He feels happy when he gets what he wants.	4.40	4.32	4.36	F (2, 346)=0.464, p=0.629, ηp2=0.003
Feels pain: He feels pain if he is injured or damaged in some way.	4.32	4.18 ~	4.07	F (2, 346)=2.331, p=0.099, ηp2=0.013

Table 3: Results of Univariate ANOVAs including all non-excluded participants for five State probes and three Entities, Study 1.

Following Huebner, I conducted post-hoc Bonferroni tests, which revealed no significant differences. There was a marginal effect for normal human vs. human whose brain has been replaced by a CPU for *feels pain* ($p = 0.097$).

Again, I converted participant responses into categorical data and performed Chi-square tests for each of the states to compare the normal human with each transplant patient and the transplant patients with one another. No significant differences emerged between participants categorical responses for the person whose brain had been replaced by a CPU and the person whose brain had been rebuilt with other neurons. A marginal effect was detected for *feels happy*, such that participants were significantly more likely to agree with attributions of *believes* to the normal person compared with the person whose brain had been replaced by a CPU, $X^2(1, N=234) = 3.734$, $p = 0.053$. However, in this case, the chi-square test's assumption that the expected frequency for each cell should be greater than five was violated by two cells, threatening the accuracy of the statistic. In any case, this result is not predicted by embodiment, so not relevant to our discussion. Participants were significantly more likely to agree with attributions of happiness to the normal person compared with the person whose brain had been replaced by a CPU, $X^2(2, N=234) = 6.225$, $p = 0.044$. But chi-square's assumptions about the expected frequency for each cell was violated by three cells. The only significant difference that did not violate chi-square's assumptions was found between people's assessments of *feels happy* attributions to the normal person and the person whose brain had been replaced by another brain. Participants were significantly more likely to agree with happiness attributions to the normal person, $X^2(2, N=239) = 4.27$, $p = 0.039$. But this result is clearly inexplicable on embodiment grounds, since both entities are fully biologically embodied (see figure 2 for patterns of attribution agreement).

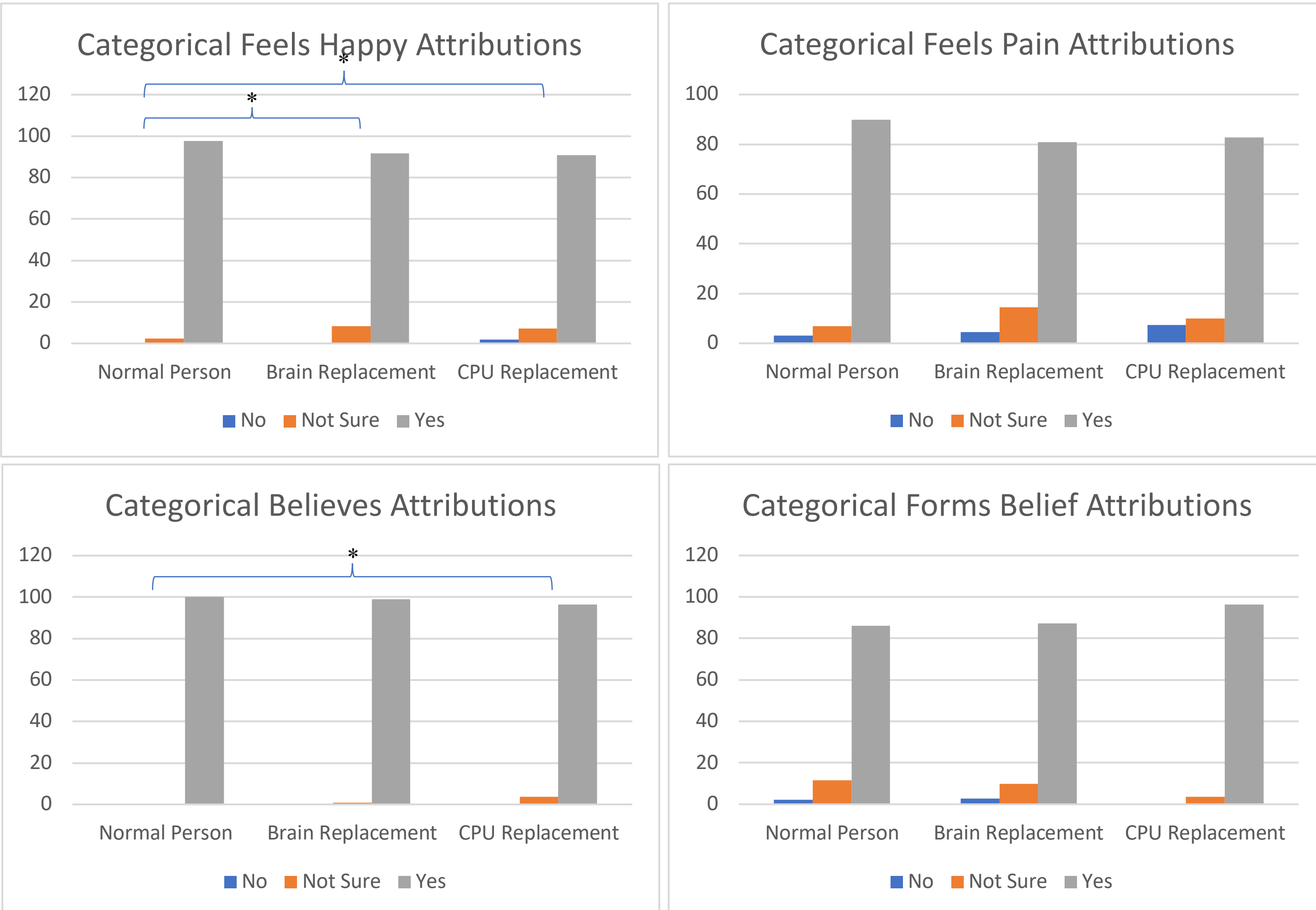

Figure 1: Categorical assessments of appropriateness for different state attributions, expressed as a percentile of 100% of answers for entity (excluding participants who regarded the entity they rated as functionally distinct from a person).

Having shown in the previous analyses a tendency for people to regard a human whose brain has been replaced by a CPU, as well as a human whose brain has been replaced by another brain, as functionally distinct from a normal person, I eliminated participants who did not agree with the statement "David behaves like a normal person in every way" from these analyses, in order to explore the assessments of people who actually regarded the brain-replacement patients as functionally equivalent to the normal human. Once this population was isolated, all effects for the scalar data disappeared. Effects for the categorical data also disappeared, or else were not explicable according to Embodiment. Still, there is a significant difference related to phenomenal states in the categorical data, participants were significantly more likely to agree with happiness attributions to the normal person than to the brain-replacement patient, and marginally (if somewhat suspectly) more likely to agree with happiness attributions to the normal person than to the cpu-replacement patient. Clearly, embodiment cannot explain the former result. Perhaps the difference is explained, in both cases, by the personal histories of the entities assessed. Both replacement patients have undergone significant brain surgery, whereas the normal person has not. As this is somewhat outside the scope of the current topic, I set it aside for the present discussion.[9]

V. Study #2: Functional differences in Sytsma's Prosthetic Hand Cases

Across a series of papers and numerous experiments, Justin Sytsma and co-authors have argued for a Naïve Account of folk concepts for at least some phenomenally conscious mental states. Sytsma (2012), for example, claims that ordinary people take "colors to be mind-independent qualities of worldly objects" and pains "to be mind-independent qualities of certain body parts" (188), and the bulk of Sytsma and co-authors' work on this topic has focused on defending these

[9] In a follow up study, I presented participants with the brain-replacement CPU vignette and the normal human vignette and asked them to assess the agent's capacity for feeling happy when he gets what he wants and feeling sad when things are taken away from him. While undergoing major surgery plausibly impairs one's capacity to function as a happy individual, my assumption was that it is consistent with—even conducive to—functioning as a sad one. If a functionalist account of phenomenal state attributions is correct, then, we might expect people to be less likely to agree with happiness attributions to the brain-replacement patients, but not sadness attributions. After eliminating repeat IP addresses, participants who did not complete the study, and participants who chose a response lower than 4 on the behaves probe (so were unsure or unconvinced that the entity they rated behaved like a normal person), I found that people were less likely to agree with attributions of the capacity for happiness to the brain-replacement CPU patient (4.16, on a five-point scale) than to the normal human (4.69), $t(206)=4.647$, $p<0.001$. However, they were also less likely to agree with attributions of the capacity for sadness to the brain-replacement CPU patient (4.07) than to the normal human (4.36), $t(206)=2.206$, $p=0.028$. Importantly, when this scalar data was converted, as above, to categorical data and subjected to chi-square tests, the effect for sadness went away ($p=0.453$), while the effect for happiness persisted (p=0.004).

two claims.[10] While certainly not a doctrinaire Embodiment View—and Sytsma himself would probably reject the label, since he rejects, on empirical, grounds the folk-reality of a distinction between phenomenal and non-phenomenal states—Sytsma's Naïve Account of pain does constitute a version of the embodiment hypothesis as we are construing it. After all, it maintains that information about the body of an entity factors into folk assessments of an entity's capacity for the phenomenal state of pain and that it does so independently of how functional information about the entity factors into such assessments.

In support of his Naïve Account, Sytsma has offered numerous original studies assessing attributions of pain to various sorts of entities, including normal humans, robots, people with various types of prosthetic hands, conjoined twins, and people with phantom limbs. Across these studies, Sytsma has attempted to standardize participant's assessments of the functional capacities of entities by including statements describing the behavioral capacities of contrasted entities in the same way. So, for example, in one important study, Sytsma (2012) described a woman who had lost a hand in an automobile accident and subsequently had it replaced with a prosthetic one. In one case, the prosthetic hand is robotic and described as being "built in a lab out of metal-alloys, sophisticated pneumatics, and complex computer circuits" (190). In another, the prosthetic hand is bioengineered and described as being "grown in a lab out of bone, muscle-tissue, and skin" (190). Sytsma then assessed how much participants agreed with attributions of pain to the woman, when she grabbed a hot pan with the prosthetic hand and dropped it while screaming out. Because Sytsma wants to isolate the role of these two distinct bodily compositions apart from differences in functional capacities afforded by the two sorts of hands,

[10] Arguments and experiments in support of the Naïve Account are found in Sytsma (2010); Sytsma (2012); Sytsma and Reuter (2017); Reuter (2011); Reuter, Phillips, and Sytsma (2014); Reuter, Sienhold, and Sytsma (2019); and Kim, Poth, Reuter, Sytsma (2016), among others.

he attempts to standardize the functional capacities of the two hands by stating, in both vignettes, that, "Susan is able to do all of the things with her [robotic/bioengineered] hand that she did with her old one—she's even resumed playing the piano!" But does this statement successfully control for functional inference? Or, instead, do participants infer richer functional capacities for the bioengineered hand despite this statement? In my second study, I'll demonstrate that this statement fails to control for functional inference, and that people are inferring functional differences that might well explain the effects that Sytsma attributes to embodiment. While I cannot survey all of Sytsma's studies in this chapter, the present demonstration suggests a reappraisal of those studies in favor of the Naïve Account that rely on control statements to standardize functional capacities.

Method

Participants: I aimed to recruit at least 60 participants per cell for this two-condition study. In total, 133 participants (57.9% men, age $M = 40.89$, SD = 10.1) from Amazon's Mechanical Turk completed the study. Recruitment was limited to Master Turkers for this study, and participants were paid $0.30 for participating. As in Study 1, I planned to eliminate participants who did not complete the study questions (though not the demographic probes), as well as the latter participating of any duplicate IP addresses. However, no participants met these conditions, so all were included in the analysis.

Procedure:

In this study, I wanted to provide evidence that the explicit statement intended to standardize functional capacity across conditions in Sytsma's (2012) prosethetic hand study, that "Susan is able to do all of the things with her [robotic/bioengineered] hand that she did with her old one," fails to control for participants' inferences to functional information. To do this, in this mixed-model, 2x2 study, I presented participants in each of two conditions with the first paragraph of either Sytsma's robotic hand or his bioengineered hand vignette, as follows:

> Susan is a normal adult female except that she lost her right hand in an automobile accident when she was 21. She has recently had an experimental procedure done that replaced her missing hand with a [robotic/bioengineered] one. The new hand was [built in a lab out of metal-alloys, sophisticated pneumatics, and complex computer circuits/grown in a lab out of bone, muscle tissue, and skin]. It is [hard and metallic/soft and fleshy] and looks [different from/similar to] her old hand. [Nonetheless/What's more], Susan is able to do all of the things with her [robotic/bioengineered] hand that she did with her old one—she's even resumed playing the piano!

Participants in each condition were then asked to generate two sorts of responses, intended as independent measures of the functional capacity for their relevant prosthetic hand:

- They were asked to indicate their best guess for the year during which Susan's experimental procedure is likely to take place on a scale running from the year to 2000 to 2500, graduated in yearly increments.
- They were asked to rate how technologically advanced Susan's prosthetic hand was on a scale running from 1 to 10, where 1 represented "fairly advanced, contemporary technology (for example, an iPhone)" and 10 represented "very complex, possible future technology (for example, a teleportation machine)."

My prediction was that, despite attempts at standardizing functional capacities in these vignettes through explicit statement, participants would infer that the bioengineered hand was more functionally complex, as revealed through significantly higher assessments of its technological complexity and significantly later year assessments for the vignette in which it occurred.

Results and Discussion:

A two-sided independent samples T-test revealed a significant difference between participants' assessments of the year in which the vignette involving the bioengineered hand (M = 2108) and the vignette involving the robotic hand (*M* = 2068) occurred, $t(131)=2.607$, $p = 0.01$. A two-sided independent samples T-test also revealed a significant difference between participants' assessments of how technologically advanced the bioengineered hand was (*M* = 8.37), compared with the robotic hand (*M* = 6.46), $t(131)=5.814$, $p < 0.001$. The bioengineered prosthetic hand was thought of as a very highly advanced technology available beyond the lifetime of the study participant, whereas the robotic prosthetic hand was thought of as a less advanced technology, likely available within the next several decades. Assuming that assessments of technological advance are correlated with assessments of functional complexity, the bioengineered hand is thought of as more functional complex than the robotic hand. The statement that Susan is able to do all of the things with her prosthetic hand that she was able to do with her old hand apparently fails to standardize functional complexity.

If people think of the bioengineered hand in Sytsma's (2012) studies as more functionally complex than the robotic hand, then it may be this functional difference that explains Sytsma's finding, that people are more likely to agree with attributions of pain to Susan with a fleshy hand than Susan with a metallic one. But if this alternative explanation has not been ruled out, then we can't definitively attribute the difference to the material makeup of the appendage in which the pain is supposed to reside. It is unclear, then, whether Sytsma's finding supports the Naïve Account (and thereby Embodiment) over an extreme version of Attribution Functionalism. In failing to control for or investigate inferences to function, another purported embodiment study fails to constitute evidence for the Embodiment Thesis.

VI. Attribution Functionalism Undefeated

In this chapter I've identified two experimental philosophical views about mental state attribution. Attribution Functionalism holds that people attribute those psychological states they take to be entailed by the functional capacities they attribute to an entity. The Embodiment Thesis maintains that information about the body of an entity factors into folk assessments of that entity's capacity for phenomenal states and that it does so independently of functional information. An extreme version of Attribution Functionalism, which maintains that information about what an entity can do is the only information relevant to all mental state attribution, is incompatible with the Embodiment Thesis. A series of studies in experimental philosophy—the embodiment studies—purport to show that considerations of the material constitution of an entity's body factors into folk assessments of that entity's capacity for phenomenal states and that they does so independently of functional information. However, as we have seen, attempts to control for inferences to functional information in at least some of these studies fail. Participants are inferring functional differences that might explain the results of the embodiment studies.

I have not surveyed all the embodiment studies in this chapter. There may be some in that large and growing literature that are more mindful of the possibility of inference to function than the two on which I have focused.[11] However, I am confident in asserting that the majority of studies in this literature rely on statements in vignettes to control for inferences to distinct functional capacity.[12] I know of no embodiment study that includes probes of inferences to functional difference, much less one that attempts to demonstrate that the effects it reports are not mediated by such inferences. Given the likelihood, demonstrated here, of systematic inferences to confounding functional differences in embodiment studies, the evidential value of these studies to debates over phenomenal state attribution needs to be rethought.

[11] See Phelan (forthcoming) for a more thorough review of this literature.
[12] Rather, the majority of studies in this literature that attempt to control for function *at all* use statements in vignettes, since some notable studies, e.g., those in Knobe and Prinz (2008), don't attempt to control for inferences to functional differences at all, but simply assume functional equivalence in their target entities.

How do people decide what mental states to attribute to an entity? This is not the sort of question to which we should expect a single answer. Obviously, someone under the sway of a theory may with care refrain from attributing phenomenal states to entities that lack a unified, biological body. But are such considerations part of the innate mechanism that, we might suspect, people use to attribute mental states, including phenomenally conscious ones? It seems plausible that people have a mindreading mechanism, inclusive of phenomenal state attributions, and that we might be able to identify a domain of stimuli that mechanism is responsive to.[13] An extreme Attribution Functionalist maintains that this mechanism's input domain includes functional information—information about what an entity can and is doing—and nothing more. However, inferences to functional information are complicated and not systematically understood. It is likely that they are responsive to a wide domain of information and can issue from modular transformations or general reasoning. A human hand is one of the few things that I know can function in all the ways requisite to feeling pain. A bioengineered hand is more like a human hand than a robotic one. Might I not infer that a bioengineered hand functions more like a human hand than a robotic hand does, and, if you ask, that it functions more like a human hand in more of the ways requisite to feeling pain? We are not in a position to conclude from the embodiment studies that bodily composition factors into attributions of phenomenally consciousness in any way other than by supporting inferences to function. Therefore, the extreme version of Attribution Functionalism remains a live possibility. It is the theory of mental state attribution favored by parsimony. It may also be favored by evolutionary considerations, since it is tough to see how interactions with robots, corporations, or spirits would have shaped the mind reading mechanism.

[13] For a seminal discussion of this mental mechanism as it relates to attitudinal state attribution see Carruthers (2011).

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